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Website: www.ktunotes.in

Course Code: EET201

Course Name: CIRCUITS AND NETWORKS

Max. Marks: 100

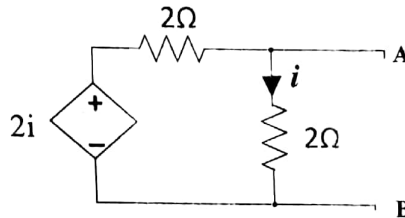
Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

1. State and explain reciprocity theorem using an example. (3)
2. Determine the Norton's equivalent circuit of the following network. (3)



3. Define time constant of a circuit. Illustrate and explain does time constant affect the charging time of the capacitor in series RC circuits connected to a DC source? (3)
4. Derive the expression for the current in a series RL circuit when connected to a DC source of voltage V , at time $t = 0$. Assume zero initial conditions. (3)
5. Obtain the transfer function of a typical series RLC circuit. Take the voltage across the capacitor as the output variable. (3)
6. Define coefficient of coupling in coupled circuits. What are its maximum and minimum values? (3)
7. A series RLC circuit with $R = 10\Omega$, $L = 2H$ and $C = 0.5F$ is connected to a 230V, variable frequency AC source. Determine the frequency of the source at which the circuit current is maximum. Also find the maximum current. (3)
8. Describe the variation of the impedance, power factor and current as a function of frequency in a series resonant circuit. (3)
9. What are h-parameters? Why are they called hybrid parameters? (3)
10. Show that the overall T-parameter matrix of two cascaded 2-port networks is the product of the T-parameter matrix of the individual networks. (3)

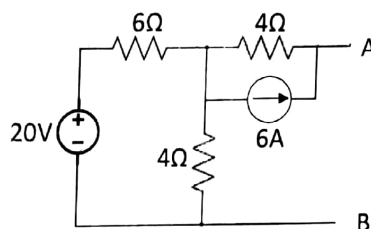
PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

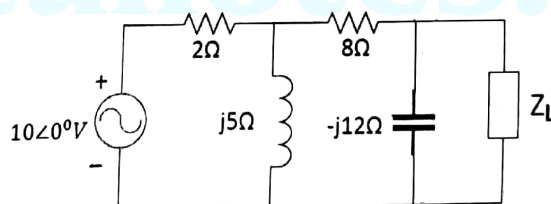
11. For the network given below,

- a) Find the Thevenin's equivalent circuit across the terminals A and B. (10)
 b) Determine the power dissipated in a 10Ω resistance when it is connected across the terminals A and B. (4)

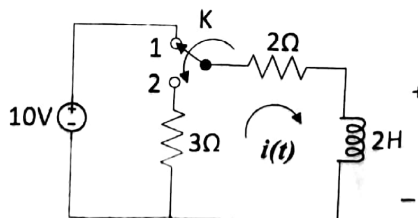


12. In the circuit given below,

- a) Determine the value of the load impedance for maximum power transferred by the source to the load. (12)
 b) Find the maximum power transferred. (2)

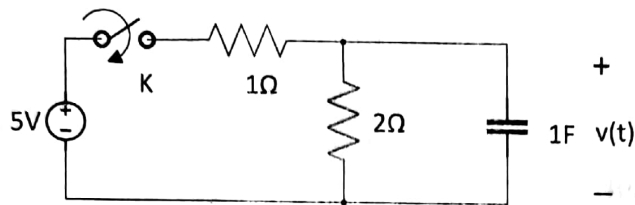
**Module 2**

13. a) In the circuit shown below, the switch was initially at position 1 and the steady state condition is reached. At $t = 0$, the switch is changed to position 2. Determine the expression for the current $i(t)$, for $t > 0$. (8)



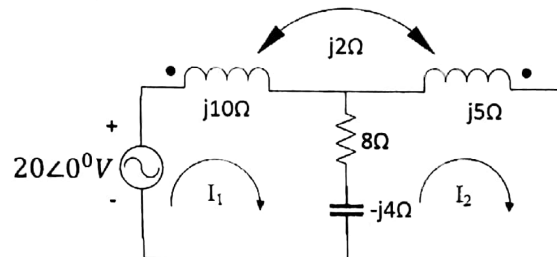
- b) A $0.25F$ capacitor with an initial voltage of $10V$ is connected across a coil of 5Ω resistance and $1H$ inductance, at time $t = 0$. Determine the expression for the current through the coil for $t > 0$. (6)

4. In the circuit given below, the switch K is closed at $t = 0$.
- a) Determine the expression for the voltage across the capacitor, $v(t)$ for $t > 0$. (10)
- b) Calculate the value of $v(t)$ at $t = 1$ seconds and its final steady state value. (4)

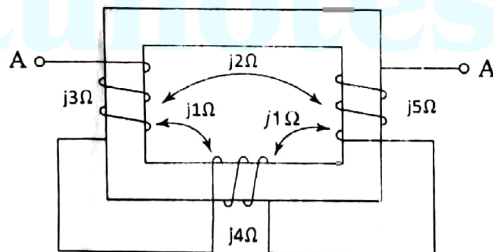


Module 3

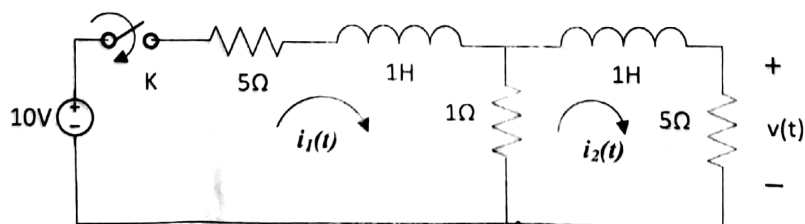
15. a) Obtain the conductively coupled equivalent circuit for the network given below. (7)
Also write the mesh equations for the equivalent circuit.



- b) If the network given below is connected across a 230V, 50Hz AC source, determine the current supplied by the source. (7)



16. The switch K in the circuit given below is closed at $t = 0$.
- a) Determine the transformed circuit for $t > 0$. Assume zero initial conditions. (4)
- b) Find the time domain expression for the voltage $v(t)$ across the 5Ω resistor for $t > 0$. Use mesh analysis. (10)



Module 4

17. A 3-phase, 3-wire, 240V system supplies a delta-connected load in which (14)
 $Z_{AB} = 25\angle 90^\circ \Omega$, $Z_{BC} = 15\angle 30^\circ \Omega$ and $Z_{CA} = 20\angle 0^\circ \Omega$. Determine the phase currents, line currents and total power consumed by the load.
18. A 3-phase, 400V, 4-wire system has a star connected load with $Z_A = 20\Omega$, (14)
 $Z_B = 15 + j10\Omega$ and $Z_C = j5\Omega$. Find the line currents, current through the neutral conductor and the total power consumed by the load.

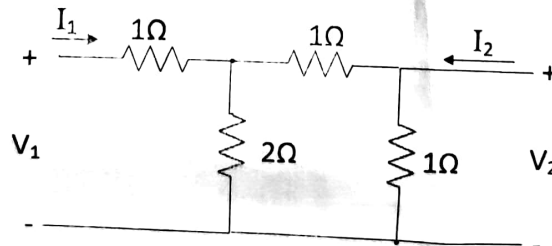
Module 5

19. a) A two port network 'A' has $[Z]_A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and another two port network 'B' (7)
has $[Y]_B = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. If the two networks A and B are connected in series, find the Z parameters of the overall network.
- b) Find the equivalent π network of a two port network represented by the (7)
following equations.

$$V_1 = 2I_1 + I_2$$

$$V_2 = I_1 + 3I_2$$

20. a) Find the h-parameters of the following two port network. (8)



- b) Derive the conditions of symmetry and reciprocity of a two port network in terms of ABCD parameters. (6)



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Scheme for Valuation/Answer Key

Scheme of evaluation (marks in brackets) and answers of problems/key

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

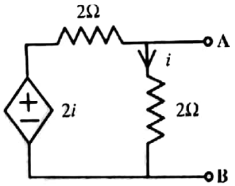
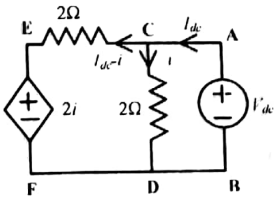
Course Code: EET201

Course Name: CIRCUITS AND NETWORKS

Max. Marks: 100

Duration: 3 Hours

PART A

Qn No.	Detailed answer key and scheme
1.	Reciprocity theorem:- Statement: In a linear, bilateral, active, single source network, the ratio of excitation to response is a constant when their positions are interchanged. (or similar statement from any standard text book) (1 mark) Explanation with an example of a linear, bilateral, active, single source network to prove reciprocity theorem. (2 marks)
2.	Norton's equivalent circuit:-  When 'A' and 'B' made short circuited, $i = 0$. Thus, $I_{AB} = I_N = 0A$ (1 mark) To find the Norton's equivalent resistance, connect an arbitrary source V_{dc} across 'A' and 'B'.  Applying KVL to loop 'ECDFE' $2i + 2(I_{dc} - i) - 2i = 0 \Rightarrow i = I_{dc}$ Applying KVL to loop 'ACDBA'

$$V_{dc} - 2i = 0 \Rightarrow V_{dc} = 2i = 2I_{dc}$$

$$\therefore \frac{V_{dc}}{I_{dc}} = Z_{AB} = Z_N = 2\Omega$$

(1)

mark)

OR

Connect an arbitrary current source of 1A across A and B and find the voltage across that current source.

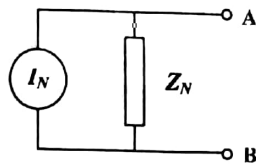
$$2(1-i) + 2i - 2i = 0$$

Hence $i = 1A$; voltage across 1A source = 2V

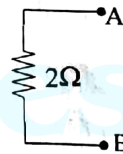
$$Z_N = \frac{2V}{1A} = 2\Omega$$

Norton's equivalent circuit:

mark)

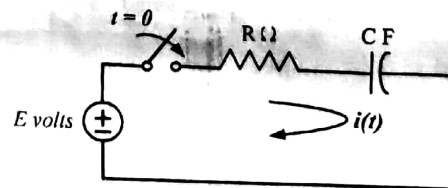


(1 mark)



3. Definition of time constant
mark)

(1)

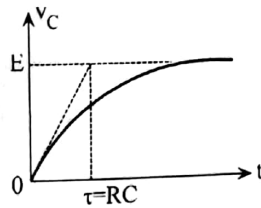


$$Ri(t) + \frac{1}{C} \int_0^{\infty} i(t) dt = E$$

$$i(t) = \frac{E}{R} e^{-\left(\frac{1}{RC}\right)t} = \frac{E}{R} e^{-\left(\frac{1}{\tau}\right)t} \quad \text{where, } \tau = RC = \text{time constant}$$

$$v_C(t) = \left(1 - e^{-\left(\frac{1}{\tau}\right)t}\right) E$$

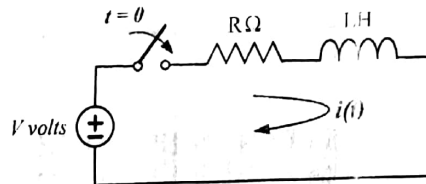
(1 mark)



As the time constant increases, charging time required to reach the steady voltage across capacitor increases or speed of response decreases. (1 mark)

(Students are not needed to derive the entire equations. Explanation with capacitor voltage equation or explanation with v_C Vs t characteristics are sufficient to award 2 marks)

4.



$$Ri(t) + L \frac{d}{dt} i(t) = V$$

(1 mark)

$$RI(s) + LsI(s) = \frac{V}{s} \quad \therefore I(s) = \frac{V}{s(R + sL)} = \frac{\left(\frac{V}{L}\right)}{s\left(s + \frac{R}{L}\right)}$$

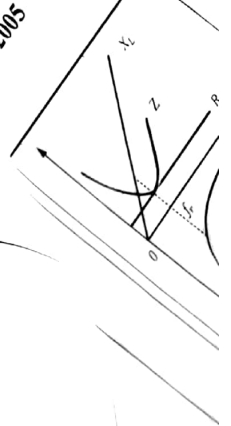
$$I(s) = \frac{\left(\frac{V}{R}\right)}{s} + \frac{\left(-\frac{V}{R}\right)}{\left(s + \frac{R}{L}\right)}$$

$$i(t) = \frac{V}{R} \left(1 - e^{-\left(\frac{Rt}{L}\right)} \right)$$

1 mark for the KVL equation, 1 mark for steps and 1 mark for expression of $i(t)$.

Steps may be considered for marks even final expression is wrong.

5. **Transfer function of RLC series circuit:-** Assume zero initial conditions.



$$TF = \frac{LT[V_{out}]}{LT[V_{in}]} = \frac{LT\left[\frac{1}{C}\int_0^{\infty} i(t)dt\right]}{LT\left[Ri(t) + L\frac{d}{dt}i(t) + \frac{1}{C}\int_0^{\infty} i(t)dt\right]} = \frac{\left[\frac{1}{sC}\right]I(s)}{\left[R + sL + \frac{1}{sC}\right]I(s)}$$

$$= \frac{\left[\frac{1}{sC}\right]}{\left[\frac{s^2LC + sRC + 1}{sC}\right]} = \frac{1}{LCs^2 + RCs + 1}$$

(3 marks) with necessary equations. 50% credit may be awarded for the steps. Full marks may be awarded for solution using any alternate method.

6. **Co-efficient of coupling (K):-**

It is defined as the fraction of the magnetic flux produced by the current in one coil that

links the other coil. $K = \frac{\phi_{21}}{\phi_1} = \frac{\phi_{12}}{\phi_2}$ (2 marks)

Maximum value of K = 1 & minimum value of K = 0:

(1 mark)

7. **Resonant frequency (f_r or ω_r) and maximum current:**

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(2 \times 0.5)}} = 0.159 \text{ Hz} \quad \text{Or} \quad \omega_r = \frac{1}{\sqrt{LC}} = 1 \text{ rad/sec} \quad (1.5 \text{ marks})$$

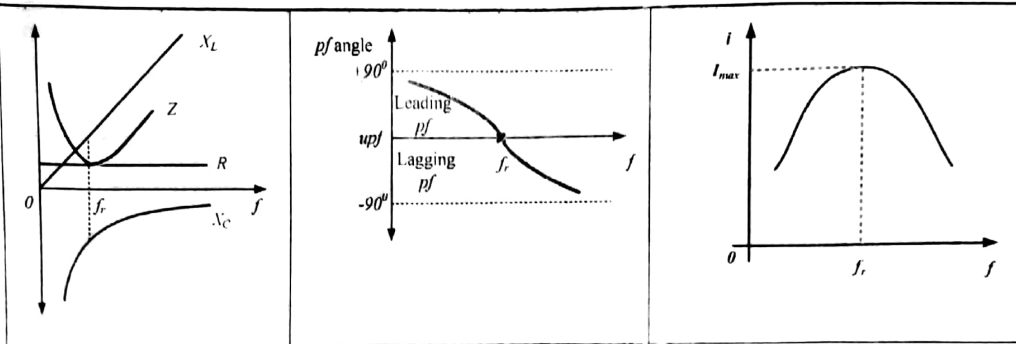
Max. current at resonance, $I_{\max} = \frac{V_s}{R} = \frac{230}{10} = 23 \text{ A}$

(1.5 marks)

8. **Impedance, power factor and current Vs frequency in series RLC circuit:-**

$f (Hz)$	$X_L (\Omega)$	$X_C (\Omega)$	$Z (\Omega)$	Phase angle	pf	current
0	0	∞	∞	-90°	leading	0
f_r	$\omega_r L$	$1/\omega_r C$	R	0°	unity	maximum
∞	∞	0	∞	$+90^\circ$	lagging	0

Impedance	Power factor	current
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- i) explanation of any one characteristics (impedance **OR** power factor **OR** current characteristics) ----- (3 marks)

OR

- ii) plot of 3 characteristics without explanation ----- (3 marks)

9. **h-parameters:-**

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \times \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0} \quad \text{SC driving point input impedance (unit : } \Omega \text{)}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0} \quad \text{SC forward current gain (no unit)}$$

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} \quad \text{OC reverse voltage gain (no unit)}$$

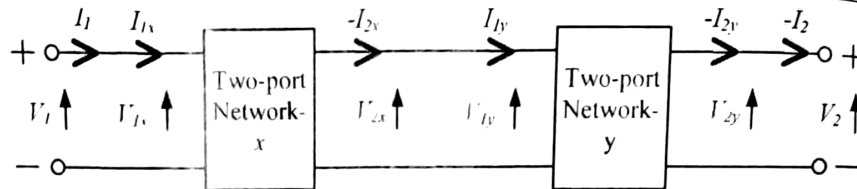
$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0} \quad \text{OC driving point output admittance (unit : } \Omega^{-1} \text{)}$$

(2)

marks)

h -parameters are also known as hybrid parameters as their units are not same [all parameters are of different nature]. It includes one impedance parameter (h_{11}), one admittance parameter (h_{22}), one voltage gain (h_{12}) and one current gain (h_{21}). (1 mark)

10. **Cascade connection of two-port networks:-**



We have,

$$\begin{aligned} V_{1x} &= A_x V_{2x} - B_x I_{2x} & V_{1y} &= A_y V_{2y} - B_y I_{2y} \\ I_{1x} &= C_x V_{2x} - D_x I_{2x} & I_{1y} &= C_y V_{2y} - D_y I_{2y} \end{aligned}$$

Now, the combined network can be written as,

For the cascade connection, from the fig.,

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} V_{1x} \\ I_{1x} \end{bmatrix} = \begin{bmatrix} A_x & B_x \\ C_x & D_x \end{bmatrix} \begin{bmatrix} V_{2x} \\ -I_{2x} \end{bmatrix}$$

$$= \begin{bmatrix} A_x & B_x \\ C_x & D_x \end{bmatrix} \begin{bmatrix} V_{1y} \\ I_{1y} \end{bmatrix}$$

$$= \begin{bmatrix} A_x & B_x \\ C_x & D_x \end{bmatrix} \left\{ \begin{bmatrix} A_y & B_y \\ C_y & D_y \end{bmatrix} \begin{bmatrix} V_{2y} \\ -I_{2y} \end{bmatrix} \right\}$$

$$= \begin{bmatrix} A_x & B_x \\ C_x & D_x \end{bmatrix} \begin{bmatrix} A_y & B_y \\ C_y & D_y \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$= \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\begin{aligned} V_1 &= V_{1x} & V_{2x} &= V_{1y} \\ V_{2y} &= V_2 & I_1 &= I_{1x} \\ -I_{2x} &= I_{1y} & -I_{2y} &= -I_2 \end{aligned}$$

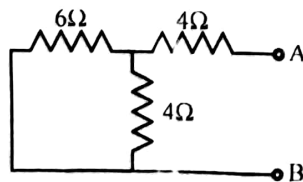
Thus, the $ABCD$ -parameters of the cascade network is given as the product of the corresponding $ABCD$ -parameters of the individual networks.

Derivation with necessary steps: 3 marks

PART B

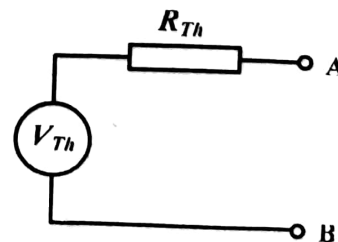
11

a) Thevenin's equivalent circuit:-



$$R_{Th} = R_{AB} = 4 + (4 \parallel 6) = 4 + \frac{4 \times 6}{4 + 6} = 4 + 2.4 = 6.4 \Omega$$

$$V_{Th} = V_{AB} = 32V$$



Thevenin's equivalent circuit

$$R_{Th} = 6.4 \Omega$$

$$V_{Th} = 32V$$

Calculation of R_{Th} : (4 marks), 50% credit for steps.

Calculation of V_{Th} : (4 marks), 50% credit for steps.

Equivalent circuit : (2 marks)

Note: Students may find V_{Th} by any suitable method.

b) Power dissipated in 10Ω resistor connected across A&B,

$$P_{10\Omega} = I^2 R = \left[\frac{32}{(6.4+10)} \right]^2 \times 10 = 38.073W$$

Or

$$P_{10\Omega} = \frac{V_{10\Omega}^2}{R} = \frac{\left[32 \times \frac{10}{(6.4+10)} \right]^2}{10} = 38.073W$$

Calculation of power: (4 marks),

50% credit for steps.

12

a) Load impedance for maximum power transfer:-

$$\begin{aligned} Z_{Th} = Z_{load} &= -j12 \parallel [8 + (j5 \parallel 2)] \\ &= 7.843 \angle -36.631^\circ \Omega = 6.294 - j4.68\Omega \end{aligned}$$

Calculation of Z_{Th} : (8 marks), 50% credit for steps.

Load impedance for maximum power transfer, $Z_L = Z_{Th}^* = 6.294 + j4.68\Omega$

Find Z_L : (4 marks)

b) Maximum power transfer:-

$$V_{Th} = V_{load} = V_{-j12\Omega} = 7.47 \angle -18.89^\circ V$$

Calculation of V_{Th} : (1 mark)

Students may find V_{Th} by any suitable method.

Maximum power,

$$P_L = \frac{V_{Th}^2}{4R_L} = \frac{7.47^2}{4 \times 6.294} = 2.216W$$

Calculation of maximum power: (1 mark)

13

Transient current:-

(a)

At position-1: steady state condition, inductor acts as SC. $I_0 = \frac{10}{2} = 5A$ (2 marks)

$$V(s) = \frac{5}{s(s+1.5)}$$

$$v(t) = \frac{10}{3} (1 - e^{-1.5t})$$

At position-2: The transient current in the circuit can be found by applying KVL to the

closed loop. $3i(t) + 2i(t) + 2 \frac{d}{dt} i(t) = 0$ (1 mark)

$$5I(s) + 2[sI(s) - I_0] = 0$$

$$\therefore I(s) = \frac{10}{(2s+5)} = \frac{5}{(s+2.5)}$$
 (3 marks) 50% credit for the steps

$$i(t) = 5e^{-2.5t}$$
 (2 marks)

13 (b) Transient current in RLC series circuit:-

The transient current in the circuit can be found by applying KVL to the closed loop.

$$\frac{1}{0.25} \int_{-\infty}^{\infty} i(t) dt + 5i(t) + 1 \frac{d}{dt} i(t) = 0$$

$$10 + \frac{1}{0.25} \int_0^t i(t) dt + 5i(t) + 1 \frac{d}{dt} i(t) = 0$$
 (2 marks)

$$\frac{4}{s} I(s) + \frac{10}{s} + 5I(s) + sI(s) = 0$$

$$I(s) = \frac{-10}{s\left(\frac{4}{s} + 5 + s\right)} = \frac{-10}{(s^2 + 5s + 4)} = \frac{-10}{(s+1)(s+4)}$$

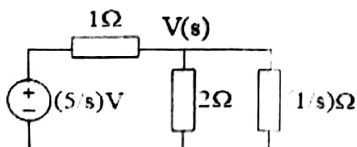
$$I(s) = \frac{-10}{(s+1)(s+4)} = \frac{\left(\frac{-10}{3}\right)}{(s+1)} + \frac{\left(\frac{10}{3}\right)}{(s+4)}$$
 (3 marks) 50% credit for the steps

$$i(t) = \frac{10}{3} (e^{-t} - e^{-4t}) \text{ A} \quad \text{OR} \quad i(t) = \frac{10}{3} (-e^{-t} + e^{-4t}) \text{ A}$$
 (1 mark)

Note: Initial voltage across capacitor may be taken as + or - 10V.

14 Voltage across capacitor:-

Transformed circuit is



Using Node analysis, $V(s) - \frac{5}{s} + \frac{V(s)}{2} + sV(s) = 0$

$$V(s) = \frac{5}{s(s+1.5)} = \frac{10/3}{s} - \frac{10/3}{s+1.5}$$

$$v(t) = \frac{10}{3}(1 - e^{-1.5t})$$

Students may find $v(t)$ by any suitable method. 50% credit for all steps. ----- (10 marks)

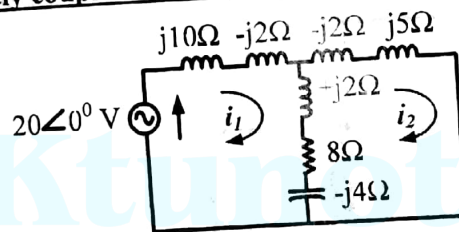
Note: Solution does not insist to draw transformed circuit. By writing differential equations, the problem can be solved.

$$v(t=1) = \frac{10}{3}(1 - e^{-1.5}) = 2.59V \text{ ----- (2 marks)}$$

$$v(t=\infty) = \frac{10}{3}(1 - e^{-\infty}) = \frac{10}{3} = 3.33V \text{ ----- (2 marks)}$$

15 **Conductively coupled equivalent circuit:-**

(a)



(3

marks)

Mesh equations:- Mesh equations by inspection method is given by,

$$\begin{bmatrix} (8 + j10 - j4) & -(8 + j2 - j4) \\ -(8 + j2 - j4) & (-j4 + 8 + j5) \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 20\angle 0^\circ \\ 0 \end{bmatrix}$$

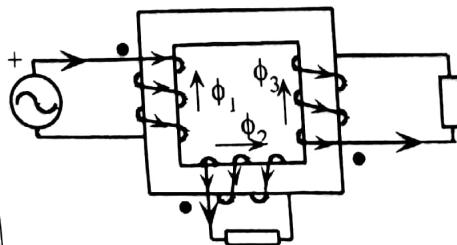
$$\begin{bmatrix} (8 + j6) & -(8 - j2) \\ -(8 - j2) & (8 + j1) \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 20\angle 0^\circ \\ 0 \end{bmatrix}$$

(2 x 2 = 4 marks)

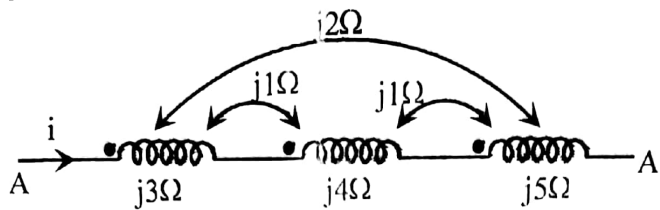
15 **Equivalent reactance and current:-**

(b) **Fixing of dots:**

Connect a voltage source across 1 coil and loads across other coils. As per the connection of source and load Fix the dots. One such connection is:



$$I_2(s) = \frac{s(s)}{s(s) + 2} = \frac{2}{s+2}$$



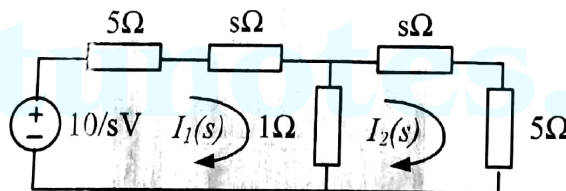
Fixing of dots (3 marks)

$$X_{eq} = X_{L_1} + X_{L_2} + X_{L_3} + 2X_{M_{12}} + 2X_{M_{23}} + 2X_{M_{31}}$$

$$= j3 + j4 + j5 + 2(j1) + 2(j1) + 2(j2) = j20\Omega$$
(2 marks)

$$\text{Current, } I = \frac{\text{Voltage}}{X_{eq}} = \frac{230 \angle 0^\circ}{20 \angle 90^\circ} = 11.5 \angle -90^\circ A = -j11.5 A$$

(2 marks)

16 **Mesh analysis:-**Transform network is,
(4 marks)

Mesh equations are,

$$\begin{bmatrix} (6+s) & -(1) \\ -(1) & (6+s) \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} \frac{10}{s} \\ 0 \end{bmatrix}$$

(2 marks) 1 mark for each mesh equations

Determination of current in mesh-2:

$$I_2(s) = \frac{\Delta Z_2}{\Delta Z}$$

$$\Delta Z = \begin{vmatrix} (6+s) & -(1) \\ -(1) & (6+s) \end{vmatrix} = (6+s)^2 - 1$$

$$\Delta Z_2 = \begin{vmatrix} (6+s) & \left(\frac{10}{s}\right) \\ -(1) & (0) \end{vmatrix} = 0 - \frac{10}{s} = -\frac{10}{s}$$

$$= s^2 + 12s + 35$$

$$\therefore I_2(s) = \frac{10}{s(s^2 + 12s + 35)} = \frac{10}{s(s+5)(s+7)}$$

(4 marks) 50% credit for the steps.



$$I_2(s) = \frac{10}{s(s+5)(s+7)} = \frac{\left(\frac{2}{7}\right)}{s} + \frac{-1}{(s+5)} + \frac{\left(\frac{5}{7}\right)}{(s+7)}$$

$$\therefore i_2(t) = \frac{2}{7} - e^{-5t} + \frac{5}{7}e^{-7t} \text{ amperes}$$

(2 marks) 50% credit for the steps.

$$V_{5\Omega}(t) = 5 \times i_2(t) = \frac{10}{7} - 5e^{-5t} + \frac{25}{7}e^{-7t} \text{ volts} \quad (2 \text{ marks})$$

17 **Delta connected load:-**

Phase sequence is not mentioned. Students may assume either ABC or ACB sequence. Also, they may take V_{AB} or V_{BC} as reference.

With phase sequence ABC:

$$V_{AB} = 240 \angle 0^\circ V \quad V_{BC} = 240 \angle -120^\circ V \quad V_{CA} = 240 \angle 120^\circ V \quad \text{----- (2 marks)}$$

$$(\text{If } V_{BC} \text{ as reference, } V_{AB} = 240 \angle 120^\circ V \quad V_{BC} = 240 \angle 0^\circ V \quad V_{CA} = 240 \angle -120^\circ V)$$

$$I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{240 \angle 0^\circ}{25 \angle 90^\circ} = 9.6 \angle -90^\circ A$$

$$I_{BC} = \frac{V_{BC}}{Z_{BC}} = \frac{240 \angle -120^\circ}{15 \angle 30^\circ} = 16 \angle -150^\circ A \quad \text{----- (3 marks)}$$

$$I_{CA} = \frac{V_{CA}}{Z_{CA}} = \frac{240 \angle 120^\circ}{20 \angle 0^\circ} = 12 \angle 120^\circ A$$

$$I_A = I_{AB} - I_{CA} = 20.873 \angle -73.3^\circ A$$

$$I_B = I_{BC} - I_{AB} = 13.95 \angle 173.4^\circ A \quad \text{----- (6 marks)}$$

$$I_C = I_{CA} - I_{BC} = 20 \angle 66.87^\circ A$$

$$\begin{aligned} P &= (V_{AB} \cdot I_{AB} \cdot \cos \phi_{AB}) + (V_{BC} \cdot I_{BC} \cdot \cos \phi_{BC}) + (V_{CA} \cdot I_{CA} \cdot \cos \phi_{CA}) \\ &= [240 \times 9.6 \times \cos(90)] + [240 \times 16 \times \cos(30)] + [240 \times 12 \times \cos(0)] \quad \text{----- (3 marks)} \\ &= 6205.537W \end{aligned}$$

OR

With phase sequence ACB:

$$V_{AB} = 240 \angle 0^\circ V \quad V_{BC} = 240 \angle 120^\circ V \quad V_{CA} = 240 \angle -120^\circ V \quad \text{----- (2 marks)}$$



$$I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{240 \angle 0^\circ}{25 \angle 90^\circ} = 9.6 \angle -90^\circ A$$

$$I_{BC} = \frac{V_{BC}}{Z_{BC}} = \frac{240 \angle 120^\circ}{15 \angle 30^\circ} = 16 \angle 90^\circ A \quad \text{----- (3 marks)}$$

$$I_{CA} = \frac{V_{CA}}{Z_{CA}} = \frac{240 \angle -120^\circ}{20 \angle 0^\circ} = 12 \angle -120^\circ A$$

$$I_A = I_{AB} - I_{CA} = 6.052 \angle 7.5^\circ A$$

$$I_B = I_{BC} - I_{AB} = 25.6 \angle 90^\circ A \quad \text{----- (6 marks)}$$

$$I_C = I_{CA} - I_{BC} = 27.066 \angle -102.8^\circ A$$

$$\begin{aligned} P &= (V_{AB} \cdot I_{AB} \cdot \cos \phi_{AB}) + (V_{BC} \cdot I_{BC} \cdot \cos \phi_{BC}) + (V_{CA} \cdot I_{CA} \cdot \cos \phi_{CA}) \\ &= [240 \times 9.6 \times \cos(90)] + [240 \times 16 \times \cos(30)] + [240 \times 12 \times \cos(0)] \quad \text{----- (3 marks)} \\ &= 6205.537W \end{aligned}$$

18

Star connected load (4-wire):-

Phase sequence is not mentioned. Students may assume either ABC or ACB sequence.

With phase sequence ABC:

$$V_A = 230.94 \angle 0^\circ V \quad V_B = 230.94 \angle -120^\circ V \quad V_C = 230.94 \angle 120^\circ V \quad \text{----- (2 marks)}$$

$$I_A = \frac{V_{AN}}{Z_A} = \frac{230.94 \angle 0^\circ}{20} = 11.547 \angle 0^\circ A$$

$$I_B = \frac{V_{BN}}{Z_B} = \frac{230.94 \angle -120^\circ}{15 + j10} = 12.81 \angle -153.69^\circ A \quad \text{----- (3 x 2 = 6 marks)}$$

$$I_C = \frac{V_{CN}}{Z_C} = \frac{230.94 \angle 120^\circ}{j5} = 46.188 \angle 30^\circ A$$

$$I_N = -(I_A + I_B + I_C) = 43.69 \angle -156.5^\circ A \quad \text{OR } I_N = I_A + I_B + I_C = 43.69 \angle 23.5^\circ A \quad \text{----- (3 marks)}$$

$$\begin{aligned} P &= (V_A \cdot I_A \cdot \cos \phi_A) + (V_B \cdot I_B \cdot \cos \phi_B) + (V_C \cdot I_C \cdot \cos \phi_C) \\ &= [230.94 \times 11.547 \times \cos 0] + [230.94 \times 12.81 \times \cos 33.69] + [230.94 \times 46.188 \times \cos 90] \\ &= 5128.155W = 5.128 kW \end{aligned}$$

----- (3 marks)

OR

With phase sequence ACB:

$$V_A = 230.94 \angle 0^\circ V \quad V_B = 230.94 \angle 120^\circ V \quad V_C = 230.94 \angle -120^\circ V \quad \text{----- (2 marks)}$$

$$I_A = \frac{V_{AN}}{Z_A} = \frac{230.94 \angle 0^\circ}{20} = 11.547 \angle 0^\circ A$$

$$I_B = \frac{V_{BN}}{Z_B} = \frac{230.94 \angle 120^\circ}{15 + j10} = 12.81 \angle 86.31^\circ A \quad \text{----- (3 x 2 = 6 marks)}$$

$$I_C = \frac{V_{CN}}{Z_C} = \frac{230.94 \angle -120^\circ}{j5} = 46.188 \angle 150^\circ A$$

$$I_N = -(I_A + I_B + I_C) = 45.28 \angle -52.4^\circ A \quad \text{OR } I_N = I_A + I_B + I_C = 45.28 \angle 127.6^\circ A \quad \text{----- (3 marks)}$$

$$P = (V_A I_A \cos \phi_A) + (V_B I_B \cos \phi_B) + (V_C I_C \cos \phi_C)$$

$$= [230.94 \times 11.547 \times \cos 0] + [230.94 \times 12.81 \times \cos 33.69] + [230.94 \times 46.188 \times \cos 9]$$

$$= 5128.155 W = 5.128 kW \quad \text{----- (3 marks)}$$

19 **Series connection of two-port networks:-**

(a) $[Z_A] = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad [Y_B] = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \quad (4$

$$\therefore [Z_B] = [Y_B]^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

marks)

$$[Z_T] = [Z_A] + [Z_B] = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix} \quad (3 \text{ marks})$$

19 **T to π transformation:**

b)

$$V_1 = 2I_1 + I_2$$

$$V_2 = I_1 + 3I_2$$

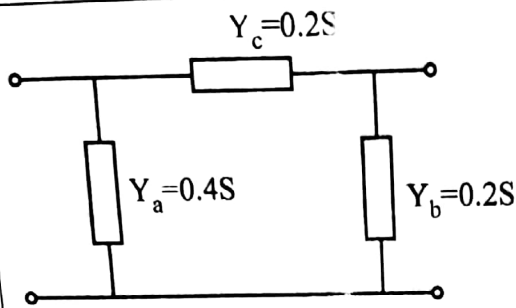
$$\therefore [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$[Y] = [Z]^{-1} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix} \quad \text{----- (3 marks)}$$

$$Y_a = Y_{11} + Y_{21} = 0.4 \Omega^{-1}$$

$$Y_b = Y_{22} + Y_{12} = 0.2 \Omega^{-1}$$

$$Y_c = -Y_{12} = -Y_{21} = 0.2 \Omega^{-1} \quad \text{----- (3 marks)}$$

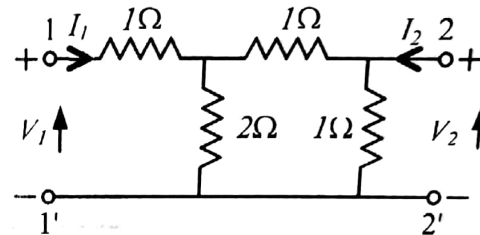


----- (1 mark)

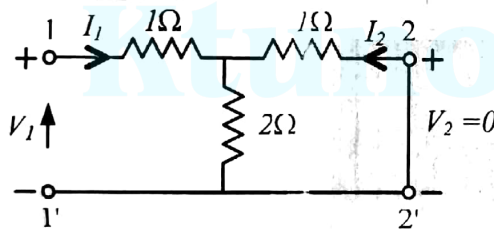
Note: Full marks may be awarded for solution using any other method.

20 h-parameters:-

(a)
$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \times \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$



Case-1 ($V_2 = 0$)



$$h_{11} = \frac{V_1}{I_1} = \frac{I_1 [1 + (2 \parallel 1)]}{I_1} = 1 + \frac{2 \times 1}{2 + 1}$$

$$= \frac{5}{3} \Omega = 1.667 \Omega$$

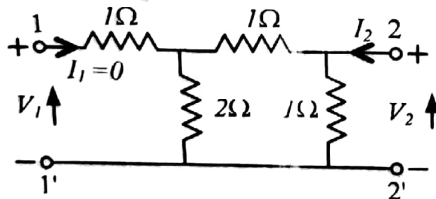
$$h_{21} = \frac{I_2}{I_1} = \frac{-I_1 \times \left(\frac{2}{2+1} \right)}{I_1} = -\frac{2}{2+1}$$

$$= -\frac{2}{3} = -0.667$$

(4 marks) 2 marks for each parameter.

50% credit for steps.

Case-2 ($I_1 = 0$)



$$h_{12} = \frac{V_1}{V_2} = \frac{V_2 \Omega}{V_2} = \frac{V_2 \times \left(\frac{2}{2+1} \right)}{V_2} = \frac{2}{2+1}$$

$$= \frac{2}{3} = 0.667$$

$$h_{22} = \frac{I_2}{V_2} = \frac{I_2}{I_2 \times (1 \parallel (1+2))} = \frac{1}{(1 \parallel 3)}$$

$$= \frac{1}{\left(\frac{1 \times 3}{1+3} \right)} = \frac{4}{3} = 1.333 \Omega^{-1}$$

(4 marks) 2 marks for each parameter.

50% credit for steps.

Note: Full marks may be awarded for solution using any other indirect method also.

Condition for symmetry and reciprocity in terms of ABCD parameters:-

Symmetry: (3 marks) 50% credit for steps.

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = \frac{AV_2 - BI_2}{CV_2 - DI_2} \bigg|_{I_2=0} = \frac{A}{C}$$

$$Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = \frac{D}{C} \quad \because I_1 = 0 = CV_2 - DI_2$$

Condition for symmetry, $Z_{11} = Z_{22}$ ie. $\frac{A}{C} = \frac{D}{C} \quad \therefore A = D$

Reciprocity: (3 marks) 50% credit for steps.

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Case-1: Excitation at port-1 (V_1), response at port - 2 (I_2) setting $V_2 = 0$

Ratio of excitation to response, $\left. \frac{V_1}{I_2} \right|_{V_2=0} = -B$

Case-2: Excitation at port-2 (V_2), response at port - 1 (I_1) setting $V_1 = 0$

$$0 = AV_2 - BI_2 \Rightarrow AV_2 = BI_2$$

$$I_1 = CV_2 - DI_2 \Rightarrow I_2 = \frac{CV_2 - I_1}{D}$$

$$\therefore AV_2 = BI_2 = B \left[\frac{CV_2 - I_1}{D} \right] = \frac{BCV_2}{D} - \frac{BI_1}{D} \Rightarrow \left(\frac{BC}{D} - A \right) V_2 = \frac{BI_1}{D}$$

Ratio of excitation to response, $\left. \frac{V_2}{I_1} \right|_{V_1=0} = \frac{B}{BC - AD}$

From case-1 and case-2, the ratio must be same for reciprocity.

$$\therefore B = \frac{B}{BC - AD} \Rightarrow BC = AD$$

Note: Full marks may be awarded for derivation using any other method
